KISHAN RAJ

Sikkim, India | +91-6207791190 | kishanraj66437@gmail.com | LinkedIn | GitHub

backed by hands-on internship experience at IIT Patna and BSPTCL. Proficient in Python, C++, and end-to-end ML pipelines, with demonstrated ability to build computer-vision systems, time-series forecasting models, and full-stack ML applications. Seeks to apply strong analytical and engineering skills to build scalable, data-driven solutions for real-world problems.

TECHNICAL SKILLS

Languages: Python, C++, SQL
AI / ML: Scikit-learn, LLM, NLP, RAG Workflow, LangChain, AI Automation, Time-Series Forecasting
Cloud & Tools: AWS (EC2, S3, ECR, IAM, CloudWatch), Git, GitHub, Docker, GitHub Actions, CI/CD
Databases: SQL, MySQL

EXPERIENCE

AI/ML Intern — Bihar State Power Transmission Company Limited (BSPTCL)

2-Month Internship

- Worked on electricity demand forecasting by merging real-world weather and electricity consumption data into a unified dataset for model training.
- Restructured consumption data at 15-minute interval granularity to improve the accuracy and resolution of short-term demand predictions.
- Applied data preprocessing and feature engineering techniques to align heterogeneous weather and consumption sources for forecasting models.

PROJECTS

Automated Face-Swapping Deep Learning Pipeline — Indian Institute of Technology, Patna

2-Month Internship | PyTorch | OpenCV | Google Colab

- Engineered an automated face-swapping deep learning pipeline supporting multi-person and multi-angle (front/left/right) identity transfer.
- Integrated pretrained CV models (MTCNN, FaceNet, InfoSwap, Ghost-SberSwap) using PyTorch and OpenCV to enhance alignment and realism.
- Deployed and validated models on Google Colab with GPU acceleration for faster inference.

CNN-Based Plant Disease Detection System

Jupyter Notebook | Python | Streamlit | NumPy | Pandas | Seaborn | Matplotlib | TensorFlow/Keras

- Designed and built an end-to-end deep learning system to classify plant diseases from leaf images using a Convolutional Neural Network (CNN) trained on a multi-class dataset.
- Shipped an interactive, user-friendly web application with Streamlit, enabling users to upload leaf images and receive real-time predictions.
- Benchmarked model performance using accuracy, precision, recall, F1-score, and confusion matrix to ensure reliable multi-class classification.

Intelligent Loan Approval System

Python | XGBoost | Scikit-learn | FastAPI | React | Docker | AWS EC2 | S3 | IAM | CloudWatch

- Built an end-to-end loan approval prediction system on 50K+ records, evaluating 5+ ML models; tuned XGBoost to 0.984 ROC-AUC and 92.6% accuracy via stratified cross-validation, with engineered risk features (DTI, LTI, PTI, net worth) and imputation, outlier capping, encoding, and scaling pipelines.
- Delivered a FastAPI + React system with real-time and batch inference and input validation; deployed the Dockerized API on AWS EC2 with model artifacts in Amazon S3 accessed via an IAM role (no hardcoded credentials).
- Configured Amazon CloudWatch for application logging and GitHub Actions CI for automated testing (pytest, Selenium) and Docker image builds.

EDUCATION

B.Tech - Computer Science & Engineering (AI & ML) 2023 - Present
Sikkim Manipal Institute of Technology | CGPA: 7.2 / 10
Relevant Coursework: Machine Learning, Deep Learning, Database Management Systems, Operating Systems, Data Structures & Algorithms, Computer Networks
Class XII — 72%

ACHIEVEMENTS

- Selected for the PwC Launchpad Program, gaining exposure to technical and professional development.
- Selected for a 2-month internship at IIT Patna, gaining practical exposure to research and real-world technical applications.